B.Tech Mathematics Mini Project (C1)

TEAM EPIDEMICS

Epidemic spreads over n weeks

TEAM LEAD

Kalingi Kesava

Roll No: 26B21A4239

TEAM MEMBERS

(1)

Nirjogi Diwakar

Roll No: 26B21A4240

(2)

Bandi Josith

Roll No: 26B21A4238

(3)

Pappala Sai

Roll No: 26B21A4236

(4)

Poluparthi Neeraj Kumar

Roll No: 26B21A4237

PROJECT NAME

Project

Epidemic Spreads Over n Weeks

Computing A^{52} with Cayley-Hamilton: Susceptible and Infected

Theme of the project

- **Disease spread as a Markov chain:** every week a person is either Susceptible (S) or Infected (I), and a 2x2 matrix A gives the chance of moving between the two.
- **Cayley-Hamilton theorem:** A satisfies its own characteristic polynomial, so any power A^n can be written as c0*I + c1*A, with no repeated matrix multiplication.
- **Goal:** compute A^52 (one year) and track the weekly state of the population, x(n) = x(0)*A^n.
- **Outcome:** the epidemic settles to a steady state of 75% susceptible and 25% infected, whatever the starting point.

How we get the matrix from the question

Matrix

Example: observe people at the start and end of a week (illustrative counts)

From \ To	S next	I next	Total
S now	900	100	1000
I now	150	350	500

- A[i][j] = (people moving from i to j) / (people in state i)
- Entries are probabilities, so all ≥ 0
- Each row is divided by its own total, so it sums to 1

S row: 900/1000 = 0.90, 100/1000 = 0.10 I row: 150/500 = 0.30, 350/500 = 0.70

In this project the question gives A directly. We read it as: row = state this week, column = state next week, and we check that every row sums to 1 before using it.

How we make the matrix A, step by step

Matrix

1 **Choose the states**

Two groups: S (susceptible) and I (infected). Each person is in exactly one group.

2 **Count the moves in one week**

Observe people at the start and end of a week. Count how many went S→S, S→I, I→S, I→I.

3 **Divide each row by its total**

A[i][j] = (people moving i→j) ÷ (people who started in i). Every entry is a probability.

4 **Place into the matrix and check**

Row = this week, column = next week. Each row must add up to 1.

Worked example (illustrative counts)

S row: 900 stay S, 100 become I → total 1000

I row: 150 become S, 350 stay I → total 500

Divide by the row total:

S row: 900/1000 = 0.90, 100/1000 = 0.10

I row: 150/500 = 0.30, 350/500 = 0.70

Result:

$$A = \begin{array}{c} \\ \text{from S} \\ \text{from I} \end{array} \begin{array}{c} \begin{array}{cc} \text{to S} & \text{to I} \end{array} \\ \begin{bmatrix} 0.90 & 0.10 \\ 0.30 & 0.70 \end{bmatrix} \end{array}$$

Reading it: A[S][I] = 0.10 means 10% of susceptible people become infected next week. If a question gives A directly, we skip to step 4 and just check the rows add up to 1.

The model and assumptions

Setup

$$A = \begin{bmatrix} 0.90 & 0.10 \\ 0.30 & 0.70 \end{bmatrix}$$

- S → I: 10% of susceptible people get infected each week
- I → S: 30% recover and become susceptible again (no lasting immunity)
- Rows = from-state, columns = to-state; each row sums to 1

$$x(n) = x(0) \cdot A^n, \quad x(0) = (0.95, 0.05)$$

- Assumed: constant probabilities, no births/deaths/vaccination, illustrative values
- Compared with 3×3 SIR: no Recovered state, so recovery returns to S

Step 1: Characteristic polynomial

Step 1

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - (\text{trace})\lambda + \det(A)$$

- $\text{trace}(A) = 0.90 + 0.70 = 1.6$
- $\det(A) = 0.90 \cdot 0.70 - 0.10 \cdot 0.30 = 0.63 - 0.03 = 0.60$
- $p(\lambda) = \lambda^2 - 1.6\lambda + 0.6 = (\lambda - 1)(\lambda - 0.6)$
- Eigenvalues: 1 and 0.6

$\lambda = 1$ again appears because A is stochastic.

Step 2: Cayley–Hamilton and A^n made simple

Step 2

$$p(A) = A^2 - 1.6A + 0.6I = 0 \quad \rightarrow \quad A^2 = 1.6A - 0.6I$$

Big idea: A^2 is built only from A and I, so higher powers never need big matrix multiplications.

1 Check it works

$1.6A - 0.6I = [[0.84, 0.16], [0.48, 0.52]]$, which equals $A \cdot A$ ✓

2 Every power has the same form

Since p has degree 2: $A^n = c_0 I + c_1 A$ for every n.

3 Match the eigenvalues

The same rule holds for λ: $\lambda^n = c_0 + c_1 \lambda$. Put in $\lambda = 1 \rightarrow 1 = c_0 + c_1$. Put in $\lambda = 0.6 \rightarrow 0.6^n = c_0 + 0.6 c_1$.

4 Subtract and solve

(1) − (2): $1 - 0.6^n = 0.4 c_1$, so $c_1 = (1 - 0.6^n)/0.4$ and $c_0 = 1 - c_1$.

5 Put in n = 52

$0.6^{52} \approx 2.9 \times 10^{-12}$ is almost 0, so $c_1 \approx 2.5$ and $c_0 \approx -1.5$.

Sanity check, n = 2: $c_1 = (1 - 0.36)/0.4 = 1.6$, $c_0 = -0.6$, which matches $A^2 = 1.6A - 0.6I$ ✓

Step 3: The matrix A^{52}

Step 3

$$A^{52} = c_0 I + c_1 A \approx -1.5 \cdot I + 2.5 \cdot A$$

	S	I
S	0.75	0.25
I	0.75	0.25

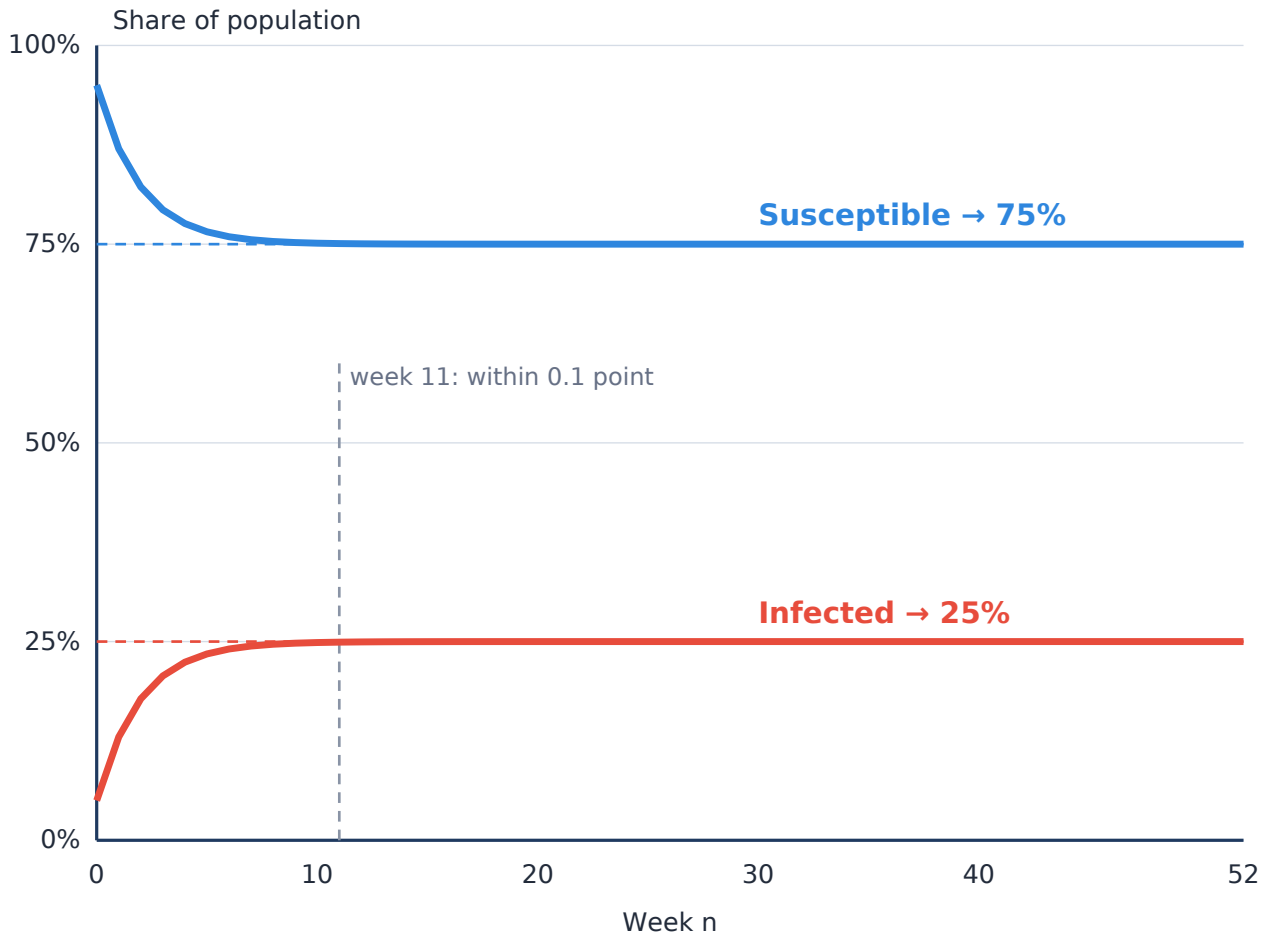
- Both rows are identical (to 11 decimals): after a year the starting state is forgotten
- Exact form: $A^n = \Pi + 0.6^n(I - \Pi)$, where every row of Π is (0.75, 0.25)
- Verified against numpy.linalg.matrix_power: error $\approx 10^{-16}$

How the epidemic evolves over 52 weeks

Step 4

$$x(n) = \pi + 0.6^n\,(x(0) - \pi)$$

Each week the gap to the steady state π = (0.75, 0.25) shrinks to 60% of what it was.



Key weeks

Week	S	I
0	95.0%	5.0%
1	87.0%	13.0%
2	82.2%	17.8%
5	76.6%	23.4%
10	75.1%	24.9%
20	75.0%	25.0%
52	75.0%	25.0%

- Infection rises quickly at first, then flattens out.
- Within 1 point of the steady state by week 6, within 0.1 point by week 11.
- After that nothing changes visibly: the starting state is forgotten.

The Python algorithm in plain language

Python

1. **Set up the numbers**
   Store A, the starting state x(0) and the number of weeks N = 52.

2. **Check A is valid**
   Each row must add up to 1, because each row is a set of probabilities.

3. **Get the polynomial**
   Take the trace and determinant of A. Their roots are the eigenvalues, 1 and 0.6.

4. **Find c_0 and c_1**
   Make $A^n = c_0 I + c_1 A$ true for both eigenvalues, then solve for c_0 and c_1.

5. **Build A^{52}**
   Multiply: c_0 times the identity plus c_1 times A. Compare with numpy as a check.

6. **Run the weeks**
   Start at x(0) and multiply by A once per week, saving every result.

7. **Find the steady state**
   Solve $\pi A = \pi$ with π adding to 1. Here $\pi = (0.75, 0.25)$.

```
import numpy as np

A  = np.array([[0.90, 0.10],
               [0.30, 0.70]])
x0 = np.array([0.95, 0.05])
N  = 52

# 2. check
assert np.allclose(A.sum(axis=1), 1)

# 3. polynomial: l^2 - tr*l + det
tr, det = np.trace(A), np.linalg.det(A)
l1, l2 = np.roots([1, -tr, det])

# 4. coefficients
c1 = (l1**N - l2**N) / (l1 - l2)
c0 = l1**N - c1 * l1

# 5. A^52 and check
AN = c0*np.eye(2) + c1*A
assert np.allclose(AN,
    np.linalg.matrix_power(A, N))

# 6. weekly states
x = [x0]
for n in range(N):
    x.append(x[-1] @ A)

# 7. steady state (2 states)
pi = np.array([A[1,0], A[0,1]])
pi = pi / pi.sum()   # (0.75, 0.25)
```

Inputs, outputs, checks and extras

Summary

INPUTS

- Transition matrix A = [[0.90, 0.10], [0.30, 0.70]]
- Start x(0) = (0.95, 0.05) as fractions of S and I
- Number of weeks N = 52
- States: Susceptible, Infected

OUTPUTS

- $p(\lambda) = \lambda^2 - 1.6\lambda + 0.6$, eigenvalues 1 and 0.6
- $c_0 \approx -1.5$, $c_1 \approx 2.5$
- $A^{52} \approx [[0.75, 0.25], [0.75, 0.25]]$
- Weekly states x(0) … x(52) with a chart
- Steady state π = (0.75, 0.25): 25% infected

CHECKS

- Every row of A adds up to 1
- $1.6A - 0.6I$ equals $A \cdot A$
- A^{52} matches numpy matrix_power (error $\approx 10^{-16}$)
- Every weekly state x(n) adds up to 1
- $\pi A = \pi$

EXTRAS

- State diagram: S→I 0.10 (infection), I→S 0.30 (recovery), stay S 0.90, stay I 0.70
- Convergence speed = second eigenvalue 0.6
- Compared with 3×3 SIR: no Recovered state, so recovery returns to S
- Assumes constant probabilities, no births, deaths or vaccination

Thank You

We sincerely thank everyone for your time,
kind attention and valuable guidance.

Team EPIDEMICS | Epidemic Spreads Over n Weeks